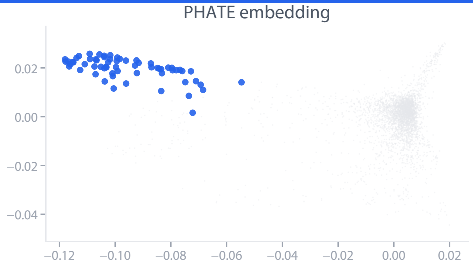
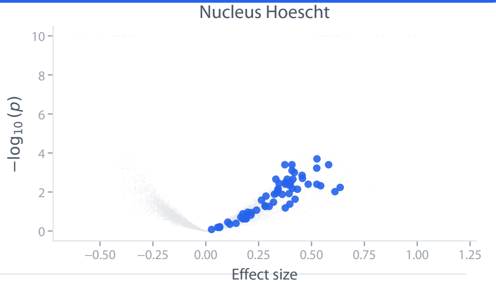
Ribosome biogenesis and large ribosomal subunit assembly

Cluster 1





Summary

This cluster shows exceptionally high confidence for ribosome biogenesis and large ribosomal subunit (60S) assembly, with >80% of genes directly involved in this pathway. The cluster contains 28 ribosomal proteins (RPL family), multiple ribosome assembly factors (GTPBP4, MDN1, GNL2, RSL24D1, MRTO4), rRNA processing enzymes (NOP2, DDX56, DDX24, DDX54, ISG20L2), and translation initiation factors (E...

Established genes

RPL30, RPL35, RPL7, RPL10A, RPL7A, RPL5, RPL37A, RPL10, RPL13A, RPL4, RPL38, RPL34, RPL19, RPL11, RPL14, RPL18A, RPL13, RPL23A, RPL31, RPL24, RPL37, RPL17, RPL9, RPLP2, RPLP1, UBA52, GTPBP4, RSL1D1, RPF1, EBNA1BP2, MDN1, GNL2, NCL, NOP2, EXOSC7, RRP1, RSL24D1, MRTO4, RRS1, RPF2, DDX56, DDX24, DDX54, ISG20L2, NLE1, EIF4G1, EIF3F, EIF3M

Novel & uncharacterized genes

NIFK *novel*

No functional annotation provided, but gene name (Nucleolar protein interacting with FHA domain of Ki-67) strongly suggests established nucleolar/ribosome function. Likely already characterized in

SPOUT1 *novel*

Recently characterized as 28S rRNA methyltransferase (2024 publication). Also has mitotic spindle functions. Role in ribosome biogenesis is now established, so priority is lower.

RPL7L1 *unc.*

No functional annotation provided, but as RPL7-like protein likely has ribosomal function. Paralog relationship suggests partial characterization exists.

BRSK2 *novel*

Serine/threonine kinase primarily studied in neuronal polarization and insulin secretion. Clustering with ribosome biogenesis factors suggests potential regulatory role, but connection is speculative.

RPL36A-HNRNPH2 *unc.*

Readthrough fusion transcript combining ribosomal protein and RNA-binding protein. Minimal functional characterization.

RPL17-C18orf32 *unc.*

Readthrough fusion transcript. Limited characterization but likely related to RPL17 function in ribosome structure.

RELB *novel*

Well-characterized NF-κB transcription factor involved in immunity and inflammation. Clustering suggests potential novel role in transcriptional regulation of ribosomal genes or nucleolar stress

DYRK1A *novel*

Dual-specificity kinase with roles in transcription, splicing, and DNA repair. Phosphorylates RNA Pol II CTD. Novel connection to ribosome biogenesis pathway not well-established, though

+ 7 more

48 established · 7 novel · 8 uncharacterized